Multimodal Evidence

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One current multimodal-evidence record with earlier speech review preserved.

Separate roles, trace a lower bound, repair certainty and plan an aligned observation.

Keep one habit: New responses remain separate from saved speech tasks; retain source scope and uncertainty.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Speech and Audio Systems. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Image Generation Models. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Iterative Prompting. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ Earlier wording, alternatives and picture transcripts

Speech and Audio Systems

Image Generation Models

Iterative Prompting

W1 · Separate evidence roles

> New tasks use separate answer fields; earlier speech/conflict tasks retain saved responses.

Current evidence and scope

Current desk record: description reports3visible pencils in openbox;closedopaquebox contentsunknown;captionclaims5total. No actual image inspected. At least3supported;no exacttotal or numericalupperbound supplied. Changed record adds verified2inside and nootherlocations,so5supported then. Keep the older speech/inventory15/50conflict separate.

Current task and new response

- Name the directly reported desk contents, the hidden information and the caption claim. Explain what source identity,time and scope would need to match before combining evidence.

My current multimodal-evidence reasoning

▼ **Review · Earlier speech task and saved response**

Reference for this exercise: open this week's student lesson, "Multimodal Evidence", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Trace the supported lower bound

> New tasks use separate answer fields; earlier speech/conflict tasks retain saved responses.

Current evidence and scope

Current desk record: description reports3visible pencils in openbox;closedopaquebox contentsunknown;captionclaims5total. No actual image inspected. At least3supported;no exacttotal or numericalupperbound supplied. Changed record adds verified2inside and nootherlocations,so5supported then. Keep the older speech/inventory15/50conflict separate.

Current task and new response

- From the supplied desk description and caption, separate direct reported observations, unknown contents and unverified claims. Explain why the supported count is a lower bound rather than an exact total. Then use the supplied changed case: opening the box reveals two pencils and there are no other desk locations.

My current multimodal-evidence reasoning

▼ **Review · Earlier speech task and saved response**

Reference for this exercise: open this week's student lesson, "Multimodal Evidence", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes. The original audio is unavailable and the assistant reports fifteen as certain.

Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair a caption-to-certainty argument

> New tasks use separate answer fields; earlier speech/conflict tasks retain saved responses.

Current evidence and scope

Current desk record: description reports3visible pencils in openbox;closedopaquebox contentsunknown;captionclaims5total. No actual image inspected. At least3supported;no exacttotal or numericalupperbound supplied. Changed record adds verified2inside and nootherlocations,so5supported then. Keep the older speech/inventory15/50conflict separate.

Current task and new response

- A caption says five pencils, but the supplied description shows only three and leaves an opaque box unobserved. Does the caption establish the hidden count or does the view disprove five? Explain.

My current multimodal-evidence reasoning

▼ **Review · Earlier speech task and saved response**

A report omits this qualification: A stage-level accuracy or authority policy does not establish every **modality** agrees or every downstream claim is supported.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Design an aligned evidence check

> New tasks use separate answer fields; earlier speech/conflict tasks retain saved responses.

Current evidence and scope

Current desk record: description reports3visible pencils in openbox;closedopaquebox contentsunknown;captionclaims5total. No actual image inspected. At least3supported;no exacttotal or numericalupperbound supplied. Changed record adds verified2inside and nootherlocations,so5supported then. Keep the older speech/inventory15/50conflict separate.

Current task and new response

- Specify one new observation or source record that could resolve the desk's unknown total. State required identity/time/location scope and the answer you could support if that evidence were supplied. Do not invent a result or reuse the independent plants case.

My current multimodal-evidence reasoning

▼ **Review · Earlier speech task and saved response**

Reference for this exercise: open this week's student lesson, "Multimodal Evidence", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Evaluate a new hidden-content description independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Evaluate only the supplied description and caption; no actual image inspection or outside inventory is provided. State which quantities are explicitly reported and which claims would need further evidence. Do not add facts about typical pot contents or unreported locations.

A description reports two visible plants and a third opaque pot whose contents cannot be seen. A caption says “three plants”. What is supported?

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Changed independent case

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Accept diagrams plus questions only when the source and topic align with the learner's task.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Earlier speech/inventory completion practice: retain this task and saved response as review.
Use the current W2 visible/hidden **evidence** task for this week's supported practice.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Speech pipelines and multimodal **evidence**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

A caption says four seedlings; the image shows three and an opaque box. Is a fourth seedling disproved? Write an **evidence**-aware response.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

An image shows three plants; caption claims a fourth outside the frame.

What can the image establish?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A chart image shows 12 plants; its caption says 21. What should a multimodal assistant report before drawing a conclusion?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 13 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

A caption says four seedlings; the image shows three and an opaque box. Is a fourth seedling disproved? Write an **evidence**-aware response.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Additional practice · Earlier room-authority task

REFERENCE · SECTION 15

Additional practice · Earlier room-authority task

> This retained exercise differs from the selected visible-plants assessment.

Earlier task and saved response

- Transcript says room E; a current written approval says room F. The task requires current approved room. Written approval is declared authoritative; trace the response and the **transcription** limit.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

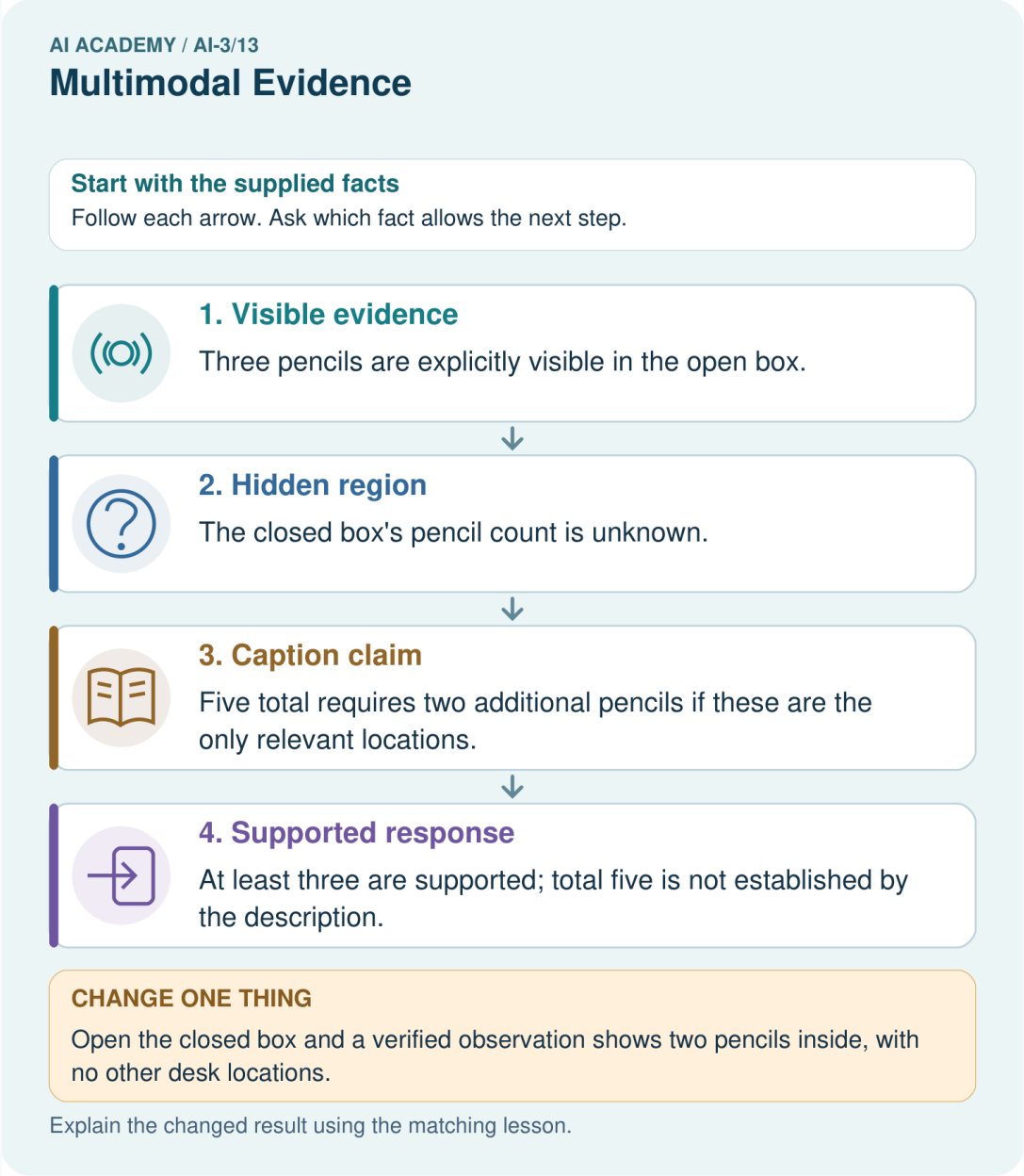
> Keep the explanation and the support you used.

Use this when helpful

A fictional Study Buddy audits a desk description and a separate caption.

Description: one open transparent box visibly contains three pencils; one closed opaque box hides its contents.

Caption: “There are five pencils on this desk.” No other **evidence** reveals the closed box.



Follow the labelled picture with the matching explanation. Open the vector diagram

Why were both “exactly three” and “exactly five” unsupported before opening the box?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Label Study Buddy claims by their visual, audio or text **evidence** and retain uncertainty when modalities omit information.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Verified Study Buddy

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Label Study Buddy claims by their visual, audio or text **evidence** and retain uncertainty when modalities omit information.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional Study Buddy audits a desk description and a separate caption.
- Which labelled step matches this detail? At least three are supported; total five is not established by the description.
- 1. Caption claim
- 2. Supported response

- 3. Hidden region
- 4. Visible **evidence**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> _____
>
> _____
>
> _____
>
> _____

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Supported response At least three are supported; total five is not established by the description.

A caption is another claim, not a way of seeing through an opaque object. Keep observable information separate from inferences.

The visible three give a lower bound under the stated desk scope; there is no supplied maximum for the closed box. Do not replace unknown contents with zero merely because none are visible.

Changed-case question

- Changed-case practice: Open the closed box and a verified **observation** shows two pencils inside, with no other desk locations. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. The total is now supported as five.
- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The total is now supported as five. 3 visible initially plus 2 newly observed gives 5 under the complete-location assumption.

Repair and save your work

Review Speech and Audio Systems (ai-3/12). Goal: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**.

Short practice: Which stage was initially wrong, and which stage was correctly operating on a wrong **input**?. Return to this week and explain what you changed.

Review Image Generation Models (ai-3/10). Goal: Explain the **mechanism** and terminology of image generation and provenance.

Short practice: What may these two grids explain, and what can they not establish about a real generator?. Return to this week and explain what you changed.

Review Iterative Prompting (ai-3/7). Goal: Explain the **mechanism** and terminology of controlled prompt experiments. Short practice: What **evidence** supports improvement here, and what remains unproven?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

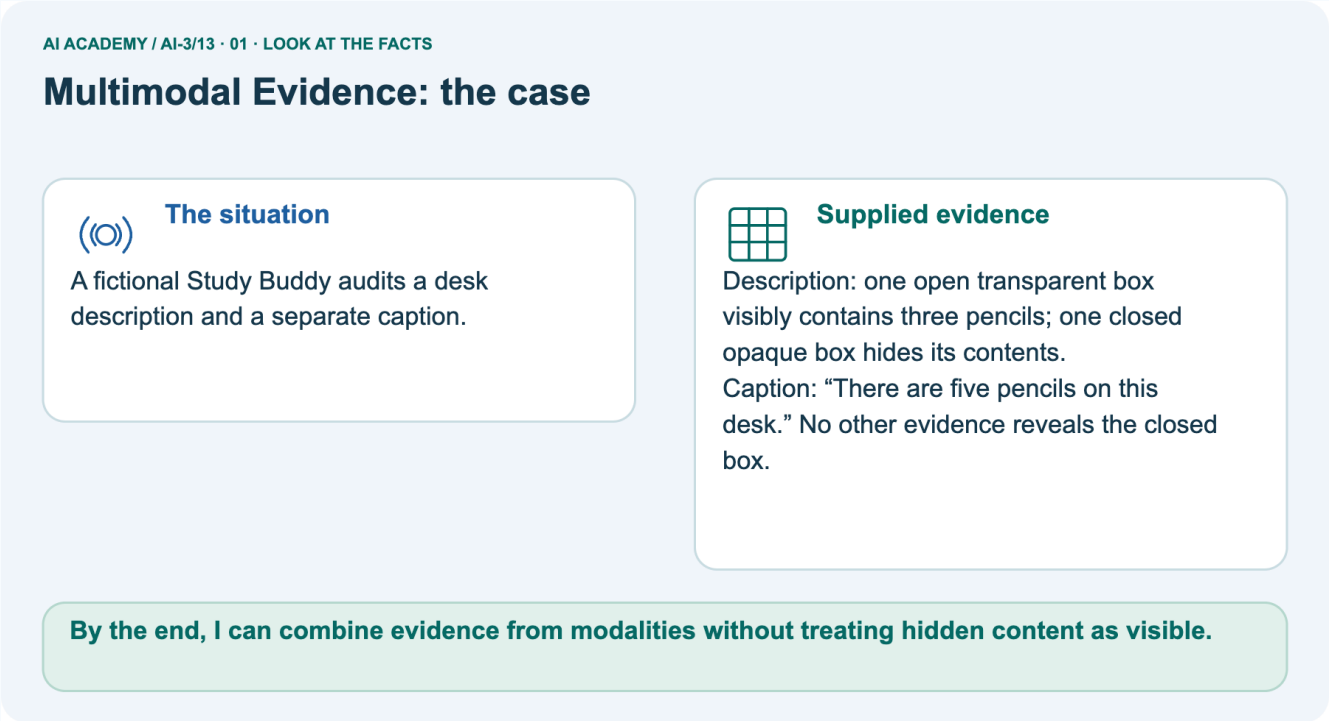
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Multimodal Evidence: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

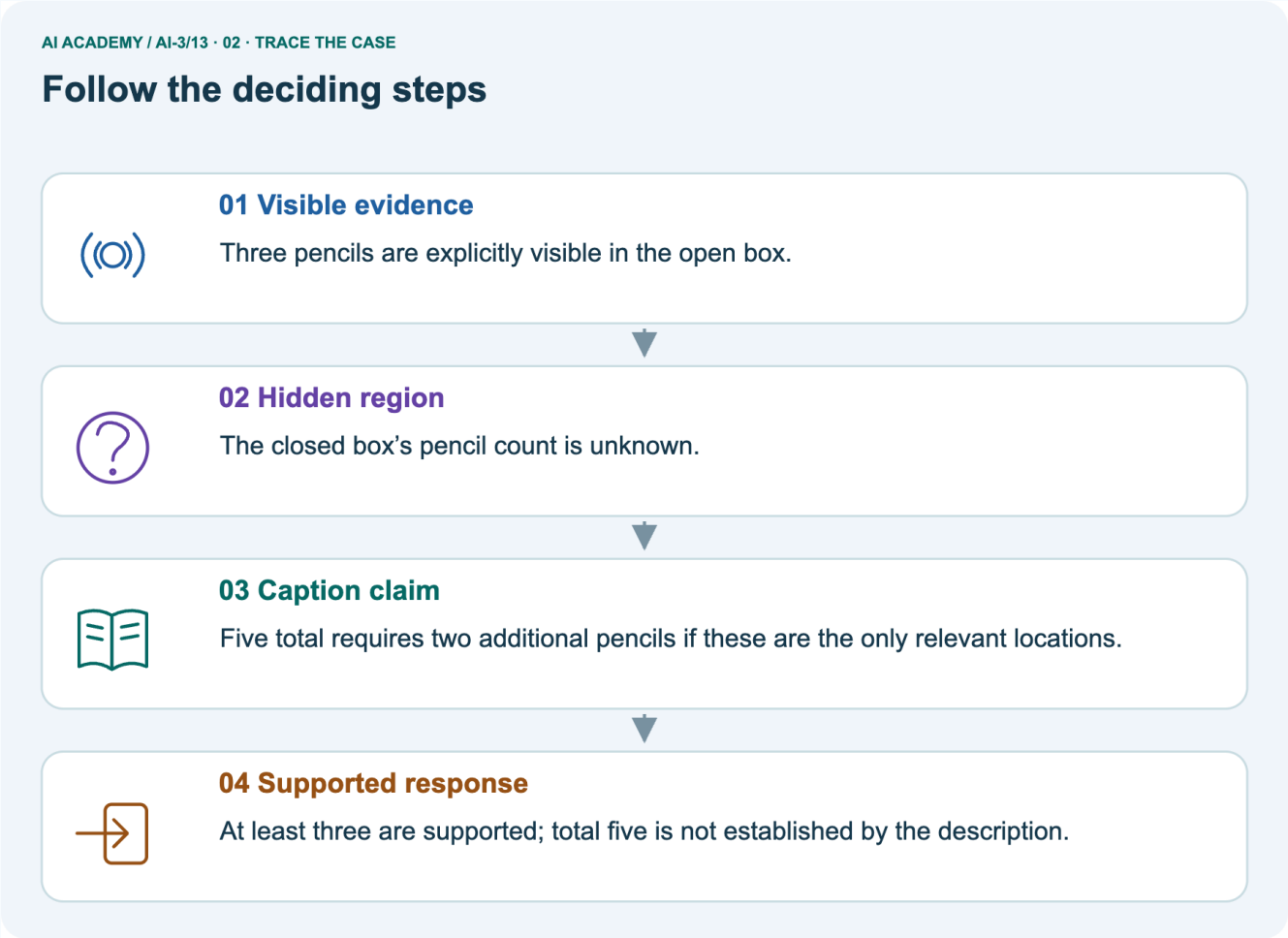
A fictional Study Buddy audits a desk description and a separate caption. Supplied facts: Description: one open transparent box visibly contains three pencils; one closed opaque box hides its contents.

Caption: “There are five pencils on this desk.” No other **evidence** reveals the closed box.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



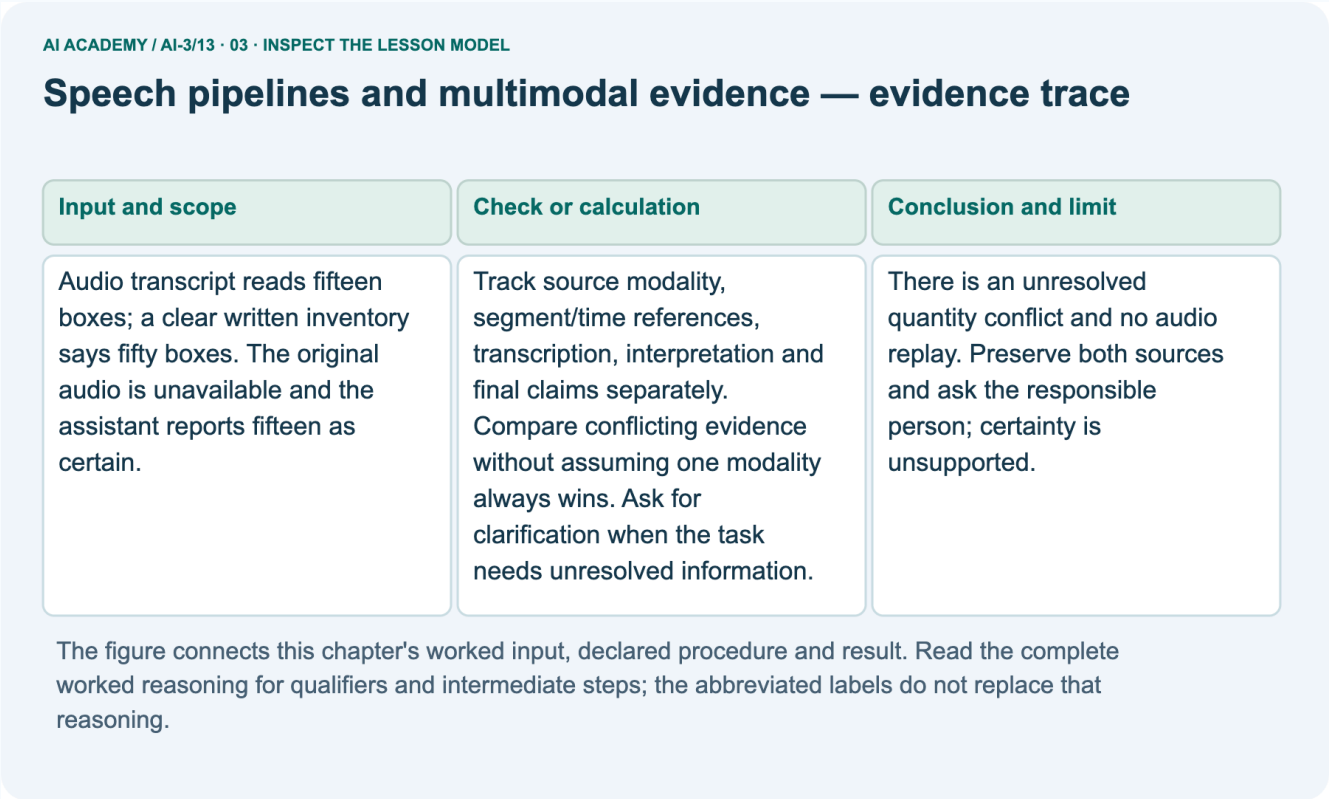
Follow the labelled picture with the matching explanation. Open the vector diagram

Visible **evidence**: Three pencils are explicitly visible in the open box. → Hidden region: The closed box's pencil count is unknown. → Caption claim: Five total requires two additional pencils if these are the only relevant locations. → Supported response: At least three are supported; total five is not established by the description.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Speech pipelines and multimodal evidence — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Speech pipelines and multimodal **evidence** — **evidence** trace. Audio transcript reads fifteen boxes; a clear written inventory says fifty boxes.

The original audio is unavailable and the assistant reports fifteen as certain. | Track source **modality**, segment/time references, **transcription**, interpretation and final claims separately.

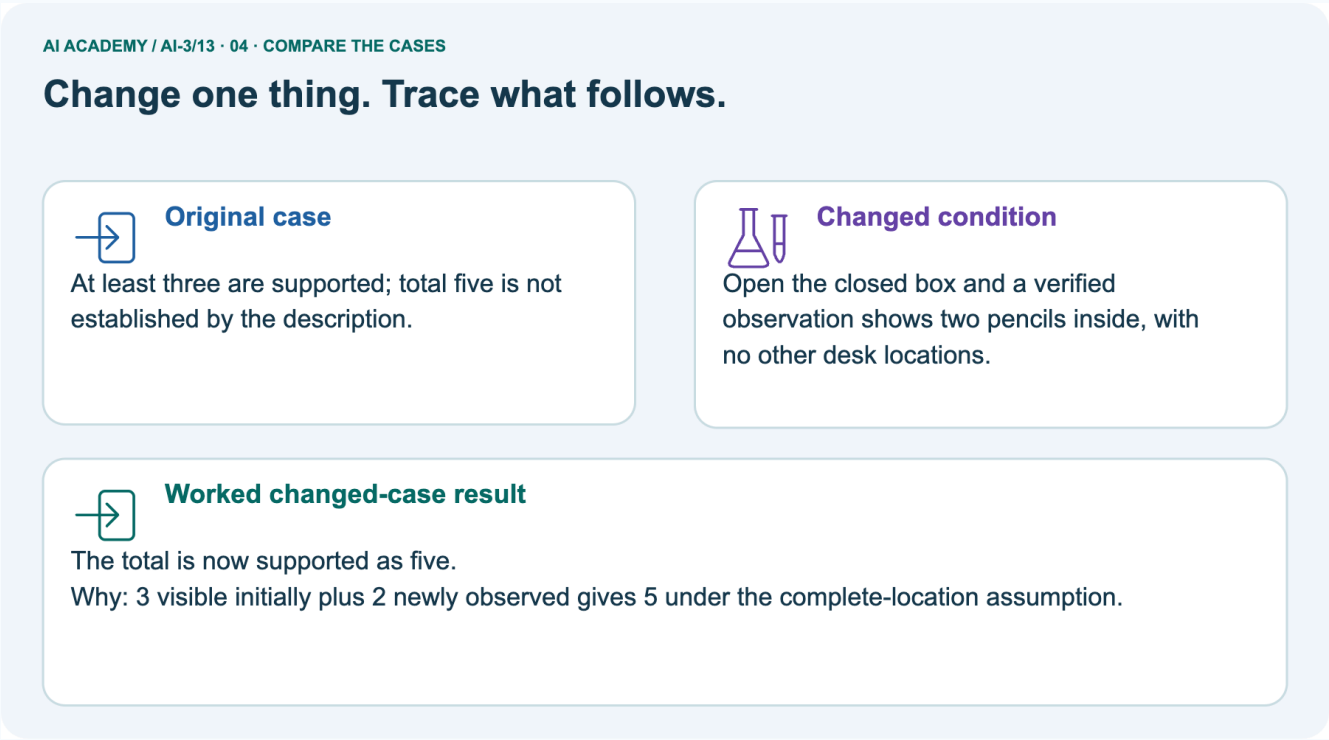
Compare **conflict**ing **evidence** without assuming one **modality** always wins. Ask for clarification when the task needs unresolved information. | There is an unresolved quantity **conflict** and no audio replay.

Preserve both sources and ask the responsible person; certainty is unsupported.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

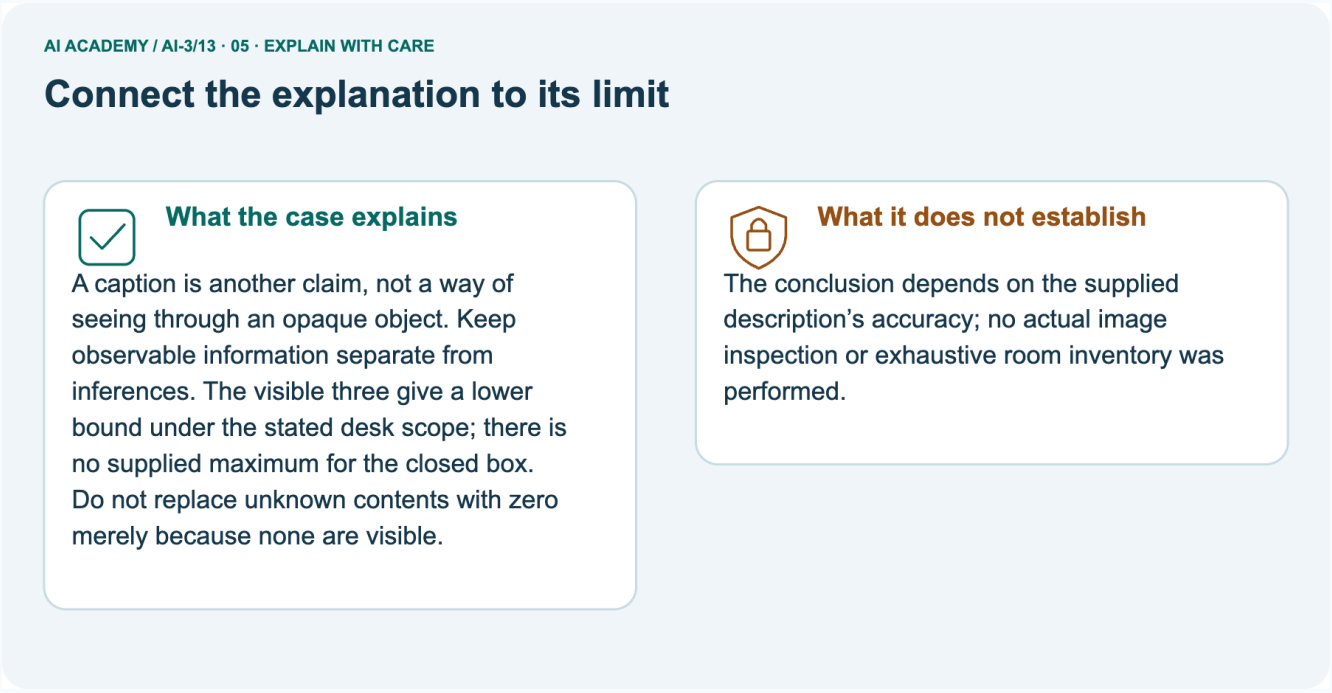
Original case: At least three are supported; total five is not established by the description. Changed condition: Open the closed box and a verified **observation** shows two pencils inside, with no other desk locations.

Expected result: The total is now supported as five. Reason: 3 visible initially plus 2 newly observed gives 5 under the complete-location assumption.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A caption is another claim, not a way of seeing through an opaque object. Keep observable information separate from inferences.

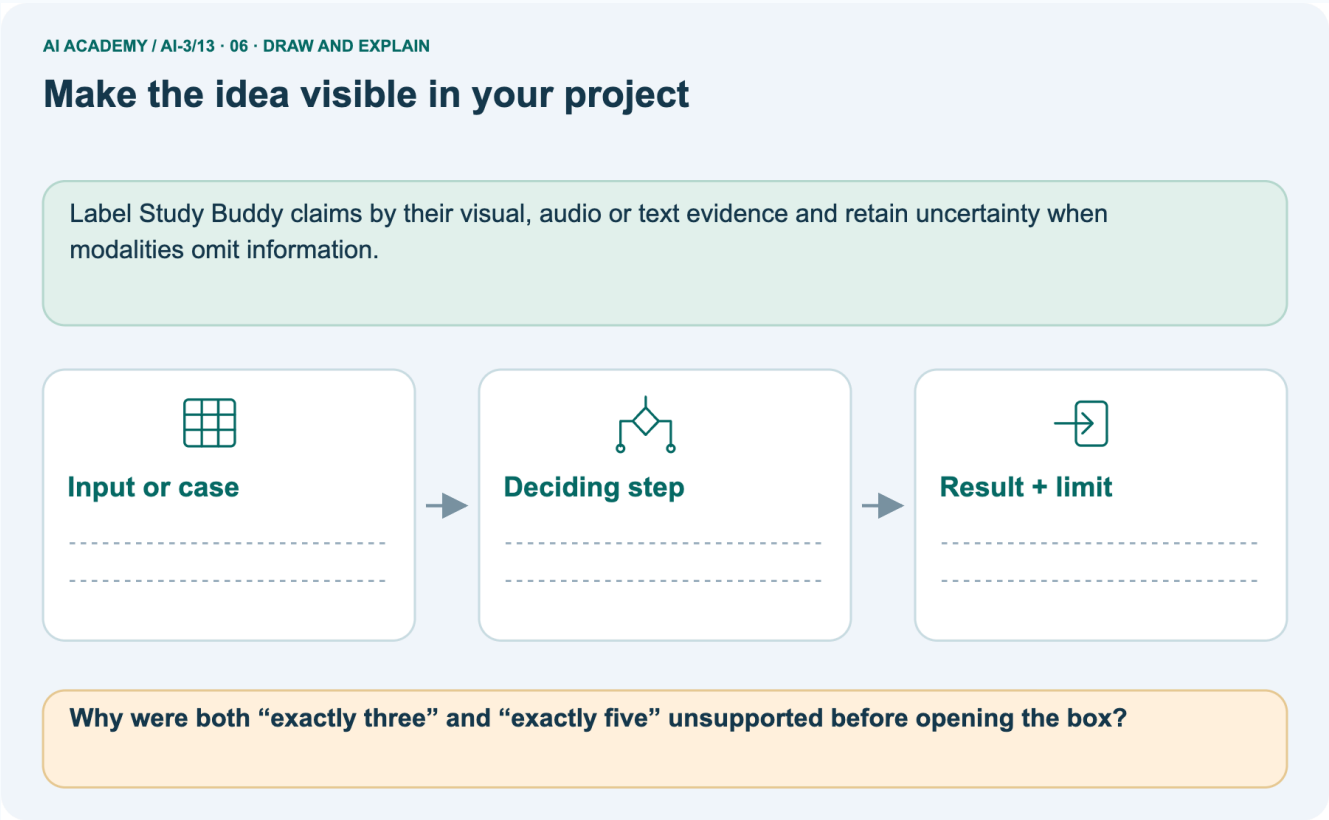
The visible three give a lower bound under the stated desk scope; there is no supplied maximum for the closed box. Do not replace unknown contents with zero merely because none are visible.

Limit: The conclusion depends on the supplied description's accuracy; no actual image inspection or exhaustive room inventory was performed.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Label Study Buddy claims by their visual, audio or text **evidence** and retain uncertainty when modalities omit information. Draw your **input**, deciding step, result and limit.

Supported question: Why were both “exactly three” and “exactly five” unsupported before opening the box?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Multimodal Evidence: the case

02. Follow the deciding steps

03. Speech pipelines and multimodal evidence — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.